Channel Self-Attention Based Multiscale Spatial-Frequency Domain Network for Oriented Object Detection in Remote Sensing Imagery

Yang Xu, *Member, IEEE*, Yushan Pan, *Graduate Student Member, IEEE*, Zebin Wu, *Senior Member, IEEE*, Zhihui Wei, *Member, IEEE*, and Tianming Zhan

***Abstract*— The detection of oriented objects in remote sensing images remains a daunting challenge due to their complex backgrounds, various sizes, and especially arbitrary orientations. However, most of the existing methods only model the structural features of the images in the spatial domain, while the horizontal convolution kernels limit the model's ability to perceive object direction information. Furthermore, the frequency features contain rich information about scale, texture, and angle, which can be a good complement to the spatial features. Inspired by this, we propose a multiscale spatial-frequency domain network (MSFN) to utilize spatial-frequency information for oriented object detection, which can be integrated into any convolutional neural network (CNN) architectures seamlessly and perform end-to-end training easily. Firstly, multiscale Haar wavelet transforms are leveraged to extract the multiscale frequency domain features from the image. Subsequently, channel alignment feature fusion module (CA-FFM) is proposed to fuse the high-level semantic features extracted by CNN with the low-level texture features extracted by the wavelet transform in multiscale. Finally, a channel self-attention (CSA)-based spatial-frequency feature perception module (SFPM) is designed to perform self-attention weighted aggregation on the fused features along the channel dimension, thereby constructing a novel spatial-frequency feature extraction backbone network for oriented object detector in remote sensing images. Experimental results on the DOTA and HRSC2016 datasets validate the effectiveness and universality of the proposed method.**

***Index Terms*— Fusion features, Haar wavelet transform, oriented object detection, remote sensing imagery, spatial-frequency domain.**

Received 27 September 2024; revised 24 October 2024; accepted 7 November 2024. Date of publication 22 November 2024; date of current version 3 December 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62471233, Grant U23B2006, Grant 62071233, and Grant 62375133; in part by Jiangsu Provincial Innovation Support Program under Grant BZ2023046; in part by Jiangsu Provincial Key Research and Development Program under Grant BE2022065-2; in part by the Key Projects of University Natural Science Fund of Jiangsu Province under Grant 23KJA520009; and in part by the Open Foundations of Jiangsu Province Engineering Research Center of Airborne Detecting and Intelligent Perceptive Technology under Grant JSECF2023-03. *(Corresponding author: Zebin Wu.)*

Yang Xu, Yushan Pan, Zebin Wu, and Zhihui Wei are with the School of Computer Science and Engineering, Nanjing University of Science and Technology (NJUST), Nanjing 210094, China (e-mail: xuyangth90@njust.edu.cn; panyushan.cn@gmail.com; zebin.wu@gmail.com; gswei@njust.edu.cn).

Tianming Zhan is with Jiangsu Key Construction Laboratory of Audit Information Engineering and the School of Information Engineering, Nanjing Audit University, Nanjing 211815, China (e-mail: ztm@nau.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3500013

I. Introduction

ORIENTED object detection in remote sensing images is always the focus and difficulty of research. Currently, most of the existing oriented object detectors rely on convolutional neural network (CNN) to extract spatial features from images. Although the presence of pooling layers gives CNN the ability to learn rotation-invariant features to a certain extent, the use of horizontal convolution kernels limits the network's ability to perceive angle information of oriented objects well. On the other hand, remote sensing images have complex backgrounds, and the texture information can significantly affect the detection results, while the frequency features of oriented objects contain rich texture and angle information that can be a supplement to spatial domain information [1].

Currently, most of the oriented object detectors focus on how to generate oriented proposals or anchors to better localize and represent object bounding boxes accurately [2], [3], [4], [5]. However, these methods often cause a massive number of parameter and high computation costs. Different from above anchor-based detectors, anchor-free detectors predict the objects directly without presetting anchors, such as MidNet [6] and SDANet [7]. Many recent methods in remote sensing imagery have focused on utilizing attention mechanisms to aggregate the contextual information of objects, highlighting the key spatial features while suppressing redundant ones, such as CFC-Net [8], FMSSD [9], LskNet [10], and PKINet [11]. Recent researches also focus on extracting the rotation-invariant features of oriented objects, aiming to enhance the ability of CNN backbone to model rotation variations [12], [13], [14]. They design a regularization constraint to align rotation-invariant features in spatial and channel dimensions. Additionally, some methods aim to optimize the loss function to alleviate discontinuity boundary problems caused by angle periodicity [15], [16], [17].

Limited by horizontal rectangular filter, ordinary CNN backbones encounter difficulties in extracting oriented object angle and texture information. Using attention mechanisms to extract high quality object features lacks effective extraction of object rotation features. Furthermore, modeling rotation-invariant features through CNN backbone requires additional parameters, thereby increasing the computational complexity of the model. Overall, the aforementioned methods primarily

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:22:32 UTC from IEEE Xplore. Restrictions apply.

focus on utilizing the spatial features of images for object detection, lacking effective exploration of frequency domain information.

By exploring the different characteristics of various image features and how they interact, we aim to make better use of frequency domain features in remote sensing images. Currently, most oriented object detectors in remote sensing imagery only extract spatial domain features of the images, lacking sufficient utilization of image frequency domain information. On the other hand, existing attention mechanisms only operate on spatial features of the images, computing similarity weights between pixel and pixel, without fully utilizing correlation information in the channel dimension. To address these issues, we propose a multiscale spatial-frequency domain network (MSFN) based on channel self-attention (CSA). This network is designed to extract the angle and texture information of oriented objects inherent in the frequency domain, thereby enhancing the detection accuracy of oriented objects in remote sensing imagery.

The feasibility of spatial-frequency feature fusion lies in the fact that there is no information loss during the mutual conversion between spatial domain images and frequency domain features. Moreover, frequency features have a more compact energy distribution, with each channel explicitly representing information from different frequency bands, including scale, angle, edges, etc. Additionally, spatial-frequency feature fusion allows for modeling object representations from multiple perspectives and extracting multidomain features, which significantly improve the accuracy of detecting oriented objects in remote sensing images [18].

Frequency domain analysis-based methods typically extract low-level texture information, which corresponds to shallow CNN features, while the features extracted by deep CNN layers have more abstract semantic information. Even if the low-level frequency features and the high-level CNN features have the same spatial scale, there is a significant difference in the channel dimension. Direct addition or concatenation operations disregard the hierarchical correlation between deep and shallow features, greatly reducing the quality of the final fused spatial-frequency features. To address the aforementioned problems, we first extract multiscale frequency features through multilevel Haar wavelet transforms. Then, we fuse the CNN features and the extracted frequency features by our proposed channel alignment feature fusion module (CA-FFM), which combines the multiscale spatial-frequency domain information with the oriented object detectors and addresses the problem of significant discrepancy in channel dimension. Finally, we integrate the proposed CA-FFM into CNN architectures seamlessly to construct the final detection framework.

The frequency domain features of an image contain rich information regarding the target's angle and texture. We can leverage these frequency domain features to recalibrate and reallocate spatial domain features or consider frequency domain information as a supplement to spatial information. By combining both, we can fully exploit the prior angular features in the data, thereby enhancing the model's ability to perceive the angular information of oriented objects and improve detection performance. On the other hand, directly adding frequency domain and spatial domain features at the channel level would lead to significant redundancy in the channel dimension. At the same time, pixel-wise addition of the two features could distort spatial information. But that is exactly what we want to get, as we aim to use frequency domain features to recalibrate and reallocate spatial features, allowing the model to perceive richer texture information and enhancing its ability to capture angular information of oriented objects.

Considering that merely expanding frequency domain features along the channel dimension via convolutional operations can result in feature redundancy and misalignment. Meanwhile, self-attention in computer vision traditionally focuses on the spatial dimension of feature maps—leading to substantial computational and memory overheads—overlooking inter-channel correlation information, we introduce an innovative CSA mechanism. It calculates self-attention along the channel dimension of feature maps to derive corresponding CSA weights, thereby guiding the model to reposition and redistribute features in the channel dimension. This method not only significantly decreases the computational and memory requirements of self-attention, but also extends the self-attention mechanism from the spatial dimension to the channel dimension, providing a novel perspective for the utilization of self-attention. Due to significant channel dimension discrepancies in spatial feature extraction, resulting in feature misalignment and redundancy after fusion, the proposed CSA module effectively addresses these issues. However, CSA operates only on single-modal features, while spatial and frequency domain features correspond to two modalities. Hence, we design the spatial-frequency feature perception module (SFPM) based on CSA, which enhances features in the channel dimension for both spatial and frequency domain modalities and can be easily extended to the case of n-modalities. Finally, the proposed CA-FFM and SFPM can be seamlessly integrated into any CNN, constructing a novel spatial-frequency feature fusion object detection network, named MSFN.

The main contributions of this article can be summarized as follows.

1) We design a multiscale spatial-frequency feature fusion network (MSFN), which leverages angle and texture information from the frequency domain to enhance oriented object detection. Additionally, to tackle the significant channel dimension discrepancy between frequency and spatial features, a CA-FFM is designed to align them.
2) This article introduces the SFPM innovatively based on CSA. While CSA operates solely on single-modal features, spatial and frequency domains represent two distinct modalities. Hence, we design the SFPM utilizing CSA to enable CSA computation for feature maps in multimodal scenarios.
3) The designed MSFN can be seamlessly integrated into various oriented object detectors. Comparative experimental results on DOTA and HRSC2016 datasets demonstrate the effectiveness and generalization ability of the proposed method.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:22:32 UTC from IEEE Xplore. Restrictions apply.

II. Related Work

A. Oriented Object Detection

Many existing oriented object detectors are two-stage detectors utilizing anchor boxes. RRPN [3] generated rotated region proposals using rotated anchor boxes, whereas RoI-Transformer [2] converts horizontal regions of interest (HRoIs) into rotated regions of interest (RRoIs), employing an RRoI learner and an RRoI aligner. In the rotation equivariant detector (ReDet) [12], the authors introduced rotation equivariant features and devised a novel module, termed as the rotated invariant regions of interest (RiRoIs) align module, for extracting rotation invariant features from rotation equivariant features.

In the realm of single-stage detectors, R^3Det [19] proposed an enhanced single-stage rotation detector, tackling feature misalignment via a coarse-to-fine-grained asymptotic regression method, and further refining detection performance through a dedicated feature refinement module. On a similar note, the align deep features network (S^2ANet) [20] introduced a novel alignment convolution mechanism to mitigate alignment discrepancies between axis-aligned convolution features and rotated object ground truth labels. Despite their promise, akin to horizontal bounding box (HBB) detectors, these approaches grappled with challenges including excessive hyper-parameters, intricate operations, and limited generalization capability.

Moreover, the challenge posed by discontinuous boundaries due to angle periodicity significantly complicates oriented object detection. Strategies such as SCRDet [21] and RSDet [22] endeavor to mitigate this issue through specialized loss functions. In CSL [23] and DCL [24], angle regression was reframed as a classification task, circumventing angle periodicity and enhancing error tolerance for neighboring angles. Concurrently, methods such as Gliding Vertex [25] and Oriented R-CNN [4] derived final object bounding boxes by identifying intersection points between inner oriented bounding boxes and outer HBBs. Nevertheless, these approaches either partially address angle prediction or rely on predefined anchor boxes, constraining further improvements in object detection performance.

B. Frequency Domain Analysis and Feature Fusion

Deep residual learning (DRL) [26] integrated frequency domain analysis techniques with CNN for JPEG image encoding. FcaNet [27] formally demonstrated that traditional global average pooling represents a special case of frequency domain feature decomposition. It extends the application of channel attention to frequency domain features and introduces a method for frequency domain channel attention in multispectral images. FsaNet [28] proposed utilizing frequency domain self-attention for image semantic segmentation. It requires only a few low-frequency coefficients as input, equivalent mathematically to spatial domain attention with a linear structure. Additionally, it proves that directional and angular information in the frequency domain can effectively relocate and redistribute spatial domain object direction and scale features. ORSIm detector [29] pioneered the construction of rotation-invariant channel features in the frequency domain, combined with original spatial-frequency features, for object detection in fused features. DFAF-Net [30] introduced the application of frequency domain attention and adaptive feature fusion in SAR-type image processing, enhancing the accuracy of object detection, particularly in datasets where conventional CNN detector encounters challenges. SpectralNET [31] proposed a fusion of wavelet transform and CNN for hyperspectral image classification, achieving promising results. GFNet, as presented by Rao et al. [32], introduced the learning of long-range spatial dependencies in the frequency domain, achieving image classification through the substitution of the SA algorithm with the 2-D discrete Fourier transform (DFF) and its inverse, leading to notable advancements in performance.

Most algorithms based on frequency domain analysis and spatial feature fusion are applied in the field of image classification. FDOL [1] proposed combining features obtained from frequency domain analysis with spatial features at multiple levels and introduced an efficient spatial-frequency feature fusion method to merge low-level frequency domain information with deep CNN semantic information. Additionally, it investigates the impact of different frequency domain analysis algorithms, such as wavelet transform and curvelet transform, on model detection performance. The effectiveness of combining frequency domain features with spatial domain features for extracting oriented object features is validated, demonstrating that spatial-frequency feature fusion can model object feature representations from multiple perspectives and can be used to extract directional features of oriented objects.

C. Attention and Self-Attention Mechanism

Since a thorough analysis of attention mechanisms is outside the purview of this work, we limit our attention to attention and self-attention mechanisms that are pertinent to computer vision. An important function of attention mechanisms in the human visual system is to reallocate resources according to importance. Over the past decade, this field has experienced significant development. Hu et al. [33] introduced SENet, demonstrating that attention mechanisms can reduce noise interference while enhancing classification accuracy. Subsequently, many articles have applied attention mechanisms to various other computer vision tasks, such as Wang et al. [34] proposed non-local networks for understanding videos, while Fu et al. [35] introduced DANet for semantic segmentation and Xie et al. [36] proposed A-SCN for point cloud processing.

The self-attention mechanism is a special case of attention mechanism. In order to capture long-range dependencies, the self-attention mechanism's primary function is to compute the similarity between features. However, as feature maps get larger, the computational complexity and memory consumption increase quadratically. To mitigate computational complexity and memory overhead, Huang et al. [37] proposed criss-cross attention, which captures global context information by sequentially considering row attention and column attention. In contrast, Li et al. [38] leveraged expectation maximization clustering to refine self-attention mechanisms, albeit with a dependency on semantic labels for optimization.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:22:32 UTC from IEEE Xplore. Restrictions apply.

In the realm of CNN, simplified approaches to self-attention mechanisms have also emerged. Some methods aim to reduce the number of keys and values. For example, in CCNet [37], each pixel only aggregated information along the vertical and horizontal paths, progressively obtaining global information through iterative processing. ISANet [39] employed short-range attention within blocks and long-range attention between blocks, achieving a reduction in computational complexity. ANN, presented by Zhu et al. [40], innovatively integrated pyramid pooling mechanisms into the branches responsible for keys and values. This architectural enhancement facilitates more comprehensive contextual information aggregation, thereby enhancing the model's performance in various computer vision tasks. ACFNet [41] introduced an attention-based feature fusion mechanism for improved object detection, while OCRNet [42] leveraged an adaptive feature refinement strategy for better semantic segmentation by capturing long-range dependencies. By means of iterative expectation maximization, EMANet [38] eliminated the query convolution and estimated a compact set of Gaussian bases. To minimize the number of keys and values, linear attention [43] created a linear attention module based on a first-order Taylor expansion. A^2Net [44] modified the sub-attention structure to implement softmax separately on query and value through second-order attention pooling, yielding a compact set of object features.

III. Proposed Method

In this section, we first present the overall structure of the proposed method (MSFN). Then, we explain how to extract the frequency domain features of the image using the Haar wavelet transform. Next, we introduce the CA-FFM and the SFPM based on CSA.

A. Overall Framework

In this article, we construct MSFN by combining multiscale spatial-frequency features. As shown in Fig. 1, we employ ResNet50 as the backbone network to extract spatial domain features. Haar wavelet transform is utilized to convert spatial domain features into frequency domain features. As the depth of the CNN and the number of wavelet transform levels increase, the spatial scale of the feature maps continuously diminishes, while the channel dimension of the CNN features steadily expands. However, the channel dimension of the frequency domain features extracted from the wavelet transform remains constant at 12. To tackle the challenge posed by the significant discrepancy in channel dimensions between spatial and frequency features, we introduce the CA-FFM. Initially, we align the frequency domain features and spatial domain features in the channel dimension. Subsequently, across various scales, spanning from low to high levels, we comprehensively fuse the two modality features. Using 1×1 convolution to expand the channel dimension to match that of spatial domain features often results in redundant information in the fused features. To address this issue, we innovatively introduce the CSA mechanism to further enhance the fused features. While conventional self-attention mechanisms typically operate on the spatial dimension of feature maps, we extend it to the channel dimension. This approach not only obtains channel-wise correlation weights to guide the model but also significantly reduces the computational complexity, enabling the designed module to be seamlessly integrated into CNNs.

B. Multiscale Frequency Domain Feature Extraction

To extract the spatial domain features, ResNet is used as the backbone to extract the basic features, which can retain the original information through convolution. For frequency domain feature extraction, we use Haar wavelet to transform spatial features into the frequency features to get more texture and angle information. Haar wavelet transform is a nonlinear wavelet transform method; as a spatial case of the Daubechies wavelet, the Haar wavelet is also known as Db1. The Haar wavelet's mother wavelet function $\psi(t)$ can be described as follows:

$$\psi(t) = \begin{cases} 1, & 0 \le t < \frac{1}{2} \\ -1, & \frac{1}{2} \le t < 1 \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

The scaling function $\phi(t)$ and filter $h[n]$ can be described as follows:

$$\phi(t) = \begin{cases} 1, & 0 \le t < 1 \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

$$h[n] = \begin{cases} \frac{1}{\sqrt{2}}, & \text{if n} = 0, 1 \\ 0, & \text{otherwise.} \end{cases} \tag{3}$$

When $n = 0$ and 1, the filter coefficients are non-zero, and the mother wavelet function can be represented using the Haar wavelet filter coefficients and its scaling function.

$$\begin{aligned} \frac{1}{\sqrt{2}}\psi\left(\left(\frac{t}{2}\right)\right) &= \sum_{n=-\infty}^{\infty} (-1)^{1-n} h[1-n]\phi(t-n) \\ &= \frac{1}{\sqrt{2}}(\phi(t-1) - \phi(t)). \end{aligned} \tag{4}$$

About Haar wavelet transform, the formula for performing the Haar wavelet transform on a matrix A is $B = \mathrm{HAH}^T$, where A is an $N \times N$ block and H is the N discrete Haar wavelet transform matrix. The inverse Haar wavelet transform is given by $A = H^T \mathrm{BH}$. To generate the matrix H, define the integer $k = 2^p + q - 1$, where $0 \le p \le n-1$. When $p = 0$, $q = 0$ or 1; when $p \ne 0$, $1 \le q \le 2^p$ [45]. Therefore, the Haar basis function can be expressed as follows:

$$h_0 z = h_{00} z = \frac{1}{\sqrt{N}}, \quad z \in [0, 1], \tag{5}$$

$$h_k(z) = h_{pq}(z) = \frac{1}{\sqrt{N}} \left\{ \begin{array}{l} 2^{\frac{p}{2}}, \frac{q-1}{2^p} \le z < \frac{q-0.5}{2^p} \\ -2^{\frac{p}{2}}, \frac{q-0.5}{2^p} \le z < \frac{q}{2^p} \\ 0, z \in [0, 1] \end{array} \right\}. \tag{6}$$

For each level of wavelet transform, it decomposes the image into a low-frequency subband (LL) and three

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:22:32 UTC from IEEE Xplore. Restrictions apply.

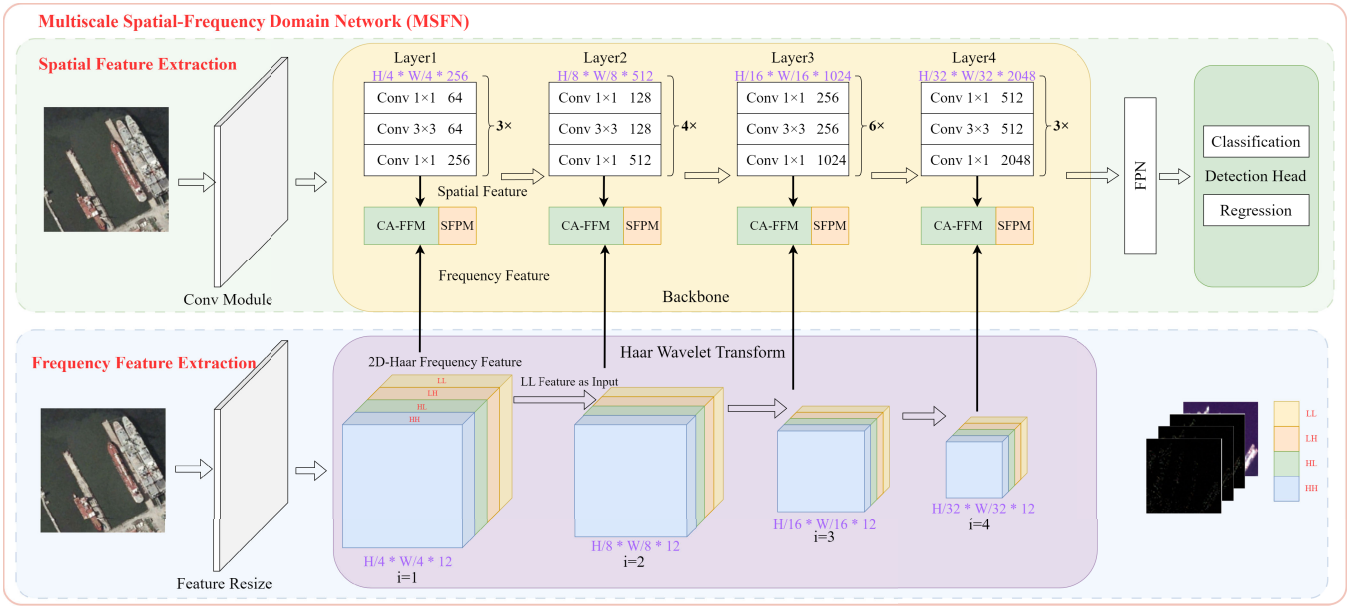


Fig. 1. Framework of our proposed model MSFN. The spatial feature extraction part utilizes ResNet to extract spatial features from images, while the frequency feature extraction part employs Haar wavelet transform to extract frequency domain features. The features extracted by both modules are channel-aligned at each layer of ResNet using our proposed CA-FFM, and finally fused through the SFPM.

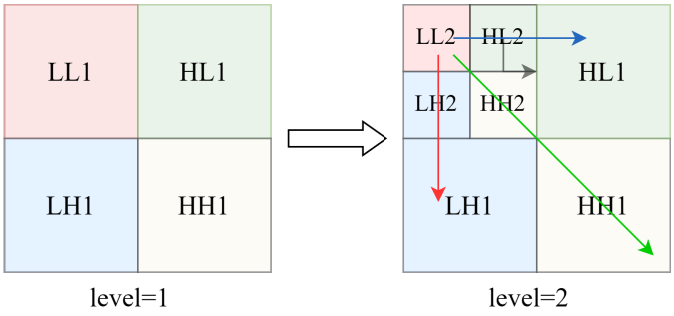


Fig. 2. Graphical illustration of Haar wavelet transform.

high-frequency subbands (LH, HL, and HH). Subsequent wavelet transform is performed based on the previous level's LL subband, and this process can be repeated to achieve i-level wavelet transform. The resolution of the image after each level of wavelet transform becomes half of the previous level. The spatial features extracted by ResNet is the same, so they can be well combined. Fig. 2 illustrates the process of Haar wavelet transform.

C. Channel Alignment Feature Fusion Module

To effectively fuse high-level spatial features and low-level frequency features, and solve the problem that the channel dimensions of the two features are significantly different to be fused directly, we design a CA-FFM. As shown in Fig. 3(a), we first extract the frequency features generated by Haar wavelet transform through 3×3 convolution. Then the frequency features and spatial features are aligned in the channel dimension by 1×1 convolution. Next, we weight the aligned frequency features in the channel dimension by squeeze and excitation (SE) channel attention [33], and then add them together to get the final fused spatial-frequency features. The fused spatial-frequency features are further weighted by SE to extract inner information in channel dimension

$$F_f = \mathrm{ReLU}\big(\mathrm{BN}\big(\mathrm{Conv3}\big(F_f\big)\big)\big) \tag{7}$$

$$\mathrm{Align}F_f = \mathrm{ReLU}\big(\mathrm{BN}\big(\mathrm{Conv1}\big(F_f\big)\big)\big) \tag{8}$$

$$\mathrm{Fusion} = \mathrm{SE}\big(\mathrm{SE}\big(\mathrm{Align}F_f\big) + F_s\big) \tag{9}$$

where, Conv3 stands for 3×3 convolution used for feature extraction, and Conv1 stands for 1×1 convolution used for channel alignment. And F_f represents frequency features, F_s represents spatial features, $\mathrm{Align}F_f$ represents the aligned features, and Fusion corresponds to the final fused features.

SE module is introduced to perform adaptive enhancement of features from the channel dimension. As is shown in Fig. 3(b), the input feature $X \in \mathbb{R}^{H \times W \times C}$ is compressed into channel descriptors by using global average pooling. Then, we use fully connected layer to excitate the descriptors and obtain the channel attention weight $W \in \mathbb{R}^{1 \times 1 \times C}$. Finally, we need to scale the weight to the original feature size and get the final attention map.

D. Channel Self-Attention

The self-attention mechanism computes the correlation between elements in a sequence to obtain corresponding weights, thereby modeling relationships and enhancing features globally pixel by pixel. The original self-attention mechanism takes 1-D vector features as input and integrates global information through three sets of different linear layers. However, in computer vision, CNN generates 2-D image features, so they are generally transformed into 1-D features through flattening operations for inter-element correlation calculation. Additionally, in computer vision, self-attention mechanisms typically operate in the spatial dimension of feature maps, computing correlations between pixels to obtain the final self-attention weights. This method obtains the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:22:32 UTC from IEEE Xplore. Restrictions apply.

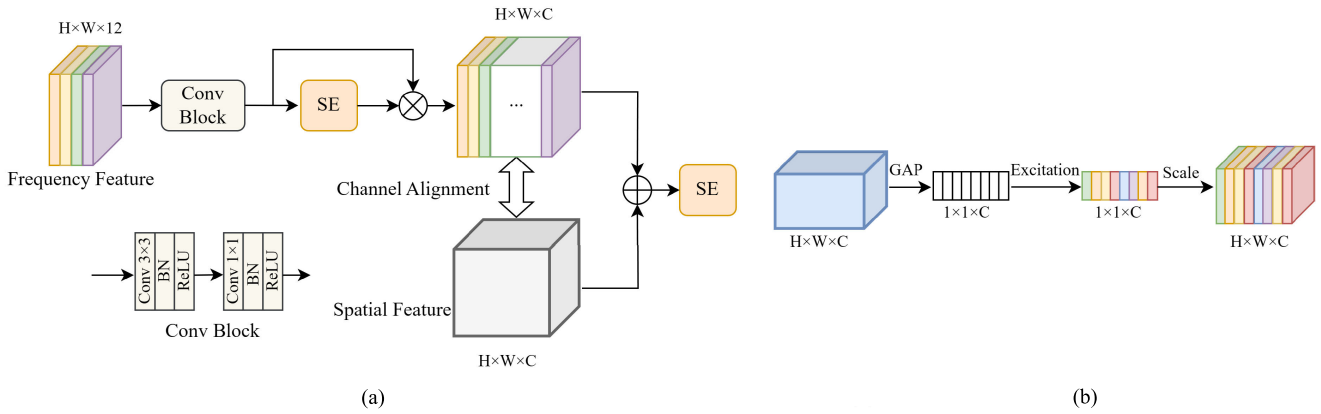


Fig. 3. (a) Proposed CA-FFM. (b) SE channel attention mechanism.

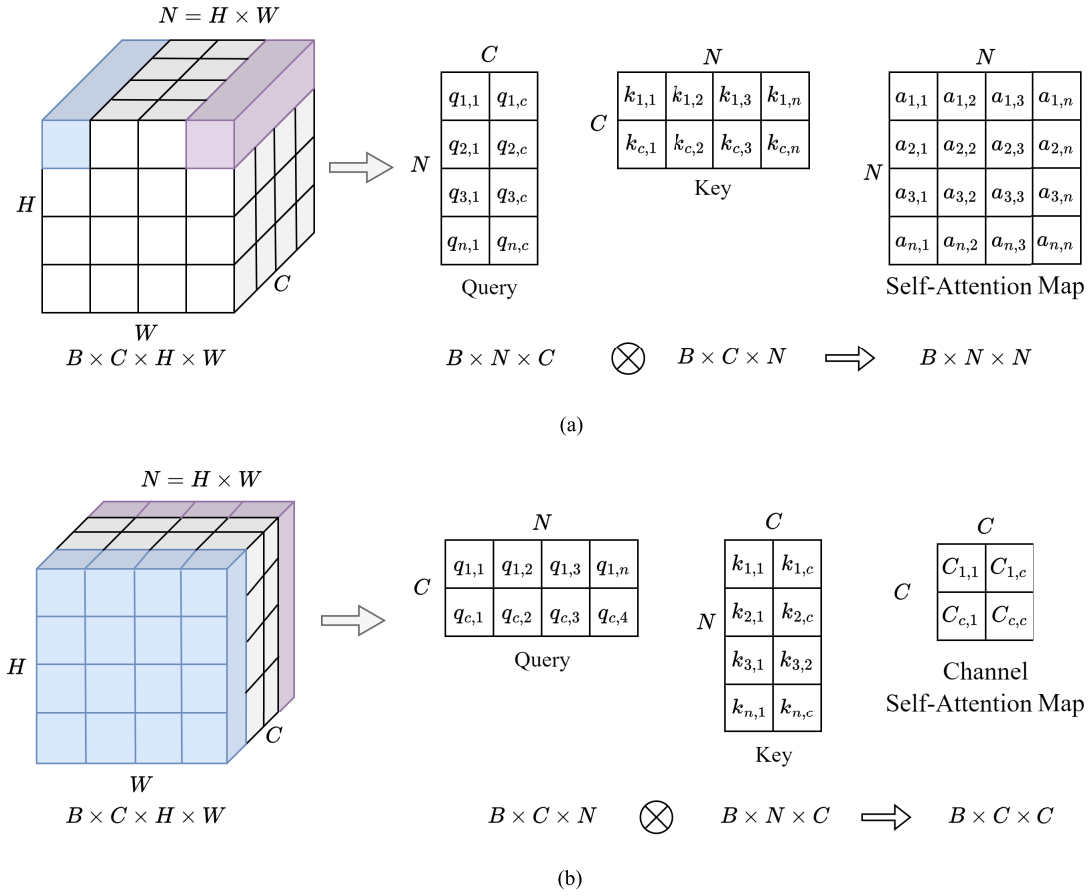


Fig. 4. Comparison diagram between traditional self-attention and the proposed CSA. (a) Traditional self-attention work on spatial feature map. (b) Proposed CSA work on channel dimension.

self-correlation between pixels under global states, neglecting the relationship between feature maps in different channels. In our proposed MSFN, a challenge in spatial-frequency feature fusion is the significant disparity in the number of channels between spatial and frequency domains, making direct two modality feature fusion difficult. Although we align frequency domain features and spatial domain features through the CA-FFM, directly adding them together results in a considerable amount of redundant information in the channel dimension. Therefore, we innovatively designed the CSA mechanism to redistribute and relocate the fused features.

In our model, the frequency domain and spatial domain features are first aligned along the channel dimension and then fused through pixel-wise addition along the same dimension. Then, the features are further fused using the SFPM. The SFPM employs CSA, as opposed to the traditional self-attention mechanism, to compute the feature maps, resulting in a channel-to-channel correlation matrix. This enables the model to recalibrate and reallocate features along the channel dimension. One of the key advantages of using CSA is that it significantly reduces the computational and memory overhead while innovatively applying the self-attention mechanism to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:22:32 UTC from IEEE Xplore. Restrictions apply.

channel-wise correlation calculations. To illustrate the relationship and collaborative patterns between frequency domain features and spatial domain features more clearly, we visualize the first five channel feature maps and are depicted in Fig. 5.

Our proposed CSA and traditional self-attention mechanism are compared in the comparison diagram shown in Fig. 4. As depicted in Fig. 4(a), traditional self-attention computes the correlation weights between two pixels (represented by blue and red squares, indicating pixels in the spatial dimension of the feature map). The original self-attention mechanism takes 1-D feature vectors as input, whereas the backbone CNN network generates 2-D feature maps. Flattening these feature maps into 1-D vectors for use in the self-attention mechanism significantly increases the computational complexity of the model. Initially, we convert the original feature map of size $B \times C \times H \times W$ into a new size of $B \times N \times C$, where $N = H \times W$ represents the number of pixels in the spatial dimension, corresponding to the size of the word vectors, and C represents the number of channels in the feature map. Consequently, the final attention feature map has a dimension of $B \times N \times N$, representing the pairwise correlations between each pixel. However, this method suffers from a large number of parameters in the final attention feature map [46]. For example, a feature map of size 128×128 yields a flattened vector of length 16 384, resulting in an attention matrix size of $16\,384 \times 16\,384$, requiring 1 GB of storage space, incurring significant time and space overheads.

The proposed CSA aims to compute the self-correlation between feature map channels. As shown in Fig. 4(b), we obtain weights between different channel feature maps (represented by blue and purple squares). After flattening the 2-D data into 1-D, we obtain a $B \times C \times N$-sized query vector, querying all pixels on the complete spatial feature map rather than a single pixel on a single channel as in traditional self-attention. The final size of the CSA feature map is $B \times C \times C$, where $C_{i,j}$ represents the CSA between the ith and jth channels. Generally, in ResNet50, the number of channels varies from 3, 64, 256, 512, 1024 to 2048. For the largest channel number of 2048, the final size of the CSA feature map is only 2048×2048, requiring only 16 MB of storage space, significantly reducing computational and storage overheads. Additionally, it expands the application of the self-attention mechanism to channel self-correlation calculation, innovatively extending its usage.

Fig. 6 illustrates the detailed structure of the proposed CSA module. From the feature maps obtained through convolution, we first pass them through three 3×3 convolutions with a stride of 2 to downscale the original feature map size to one-quarter of its original size. This process yields query, key, and value vectors, which are then flattened into one dimension. Subsequently, self-attention computation is performed on the obtained feature maps along the channel dimension, wherein matrix multiplication is followed by activation through the softmax function to obtain the corresponding CSA feature map. After obtaining the CSA feature map, it is multiplied with the value and then further processed through a 1×1 convolution layer to obtain the refined features that have undergone relocation and redistribution in the channel dimension through the CSA module.

E. Spatial-Frequency Feature Perception Module

The CSA module proposed in Section III-D can be seamlessly integrated into CNNs to refine feature maps along the channel dimension. However, CSA can only enhance single-modal features and cannot be applied to multimodal scenarios. To address the spatial-frequency feature fusion problem in this article, as shown in Fig. 7, we design the SFPM based on CSA, aiming to enhance both spatial and frequency domain features simultaneously, thus achieving multimodal CSA. The specific approach is as follows: firstly, the CSA module is used to extract CSA maps for both spatial and frequency domain features. Then, the two attention maps are added together to fuse the attention feature maps. We need to select a primary modality from the multimodal features, and then multiply the attention value feature map obtained from the primary modality with the fused feature map to obtain the final output. In this article, we perform CSA computation after fusing spatial and frequency domain features. The proposed multimodal perception module can also be extended to n-modal scenarios. The implementation method is consistent with the above. First of all, selecting a primary modality to compute the value, then calculating CSA for each modality separately, adding them together to obtain the fused weights, and finally multiplying with the value to achieve multimodal CSA.

IV. Result and Discussion

A. Dataset

HRSC2016 is a remote sensing object detection dataset comprising 1061 images and 2970 object instances. Each instance in the dataset has two types of annotations: rotated bounding box annotations and rectangular bounding box annotations. The size of each image varies from 700 to 1100 pixels. The dataset is divided into three parts: training set, test set, and validation set, containing 444, 436, and 181 images, respectively. We solely utilize the training set to train the model and validate the effectiveness of the proposed method on the test set.

The DOTAv1.0 dataset is currently one of the most commonly used datasets for oriented object detection in remote sensing images. It consists of 2806 aerial images captured from different sensors and platforms, with varying resolutions ranging from 800×800 to $20\,000 \times 20\,000$ pixels. The dataset contains 15 categories, totaling 188 282 object instances, including PL-plane, BD-baseball diamond, BR-bridge, GTF-ground field track, SV-small vehicle, LV-large vehicle, SH-ship, TC-tennis court, BC-basketball court, ST-storage tank, SBF-soccer-ball field, RA-roundabout, HA-harbor, SP-swimming pool, and HC-helicopter. Among these, there are 1411 images in the training set, 458 images in the validation set, and 937 images in the test set.

B. Evaluation Criteria

The model's detection performance is evaluated using mean average precision (mAP), with the VOC07 protocol applied

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:22:32 UTC from IEEE Xplore. Restrictions apply.

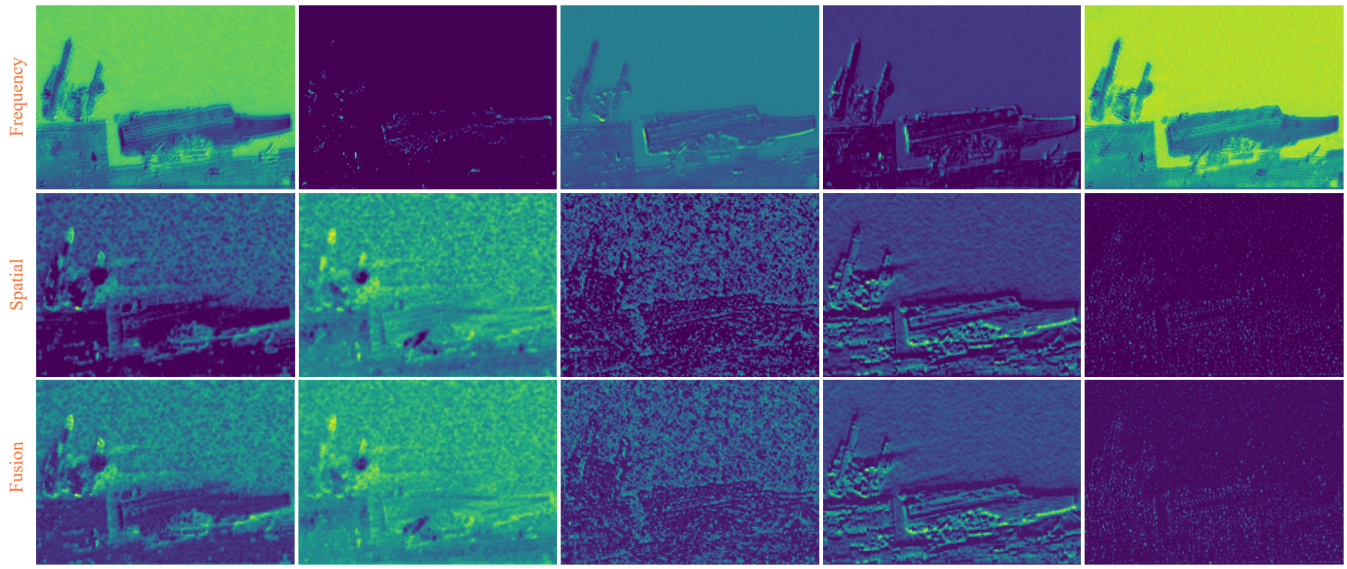


Fig. 5. Visualization of the first five channel feature maps of frequency domain features, spatial domain features, and fused features.

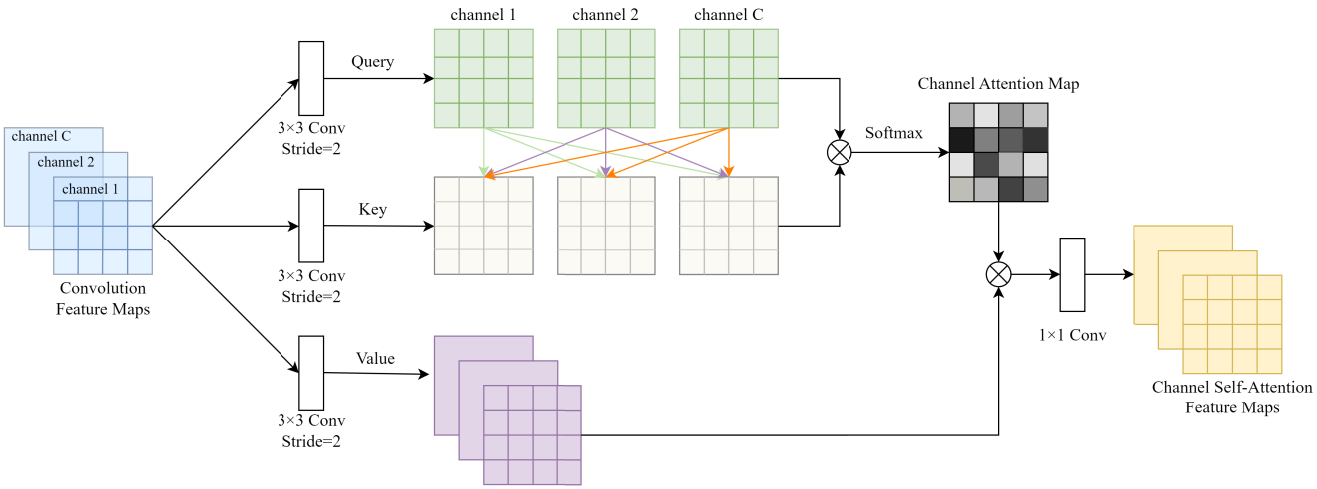


Fig. 6. Structure diagram of CSA module.

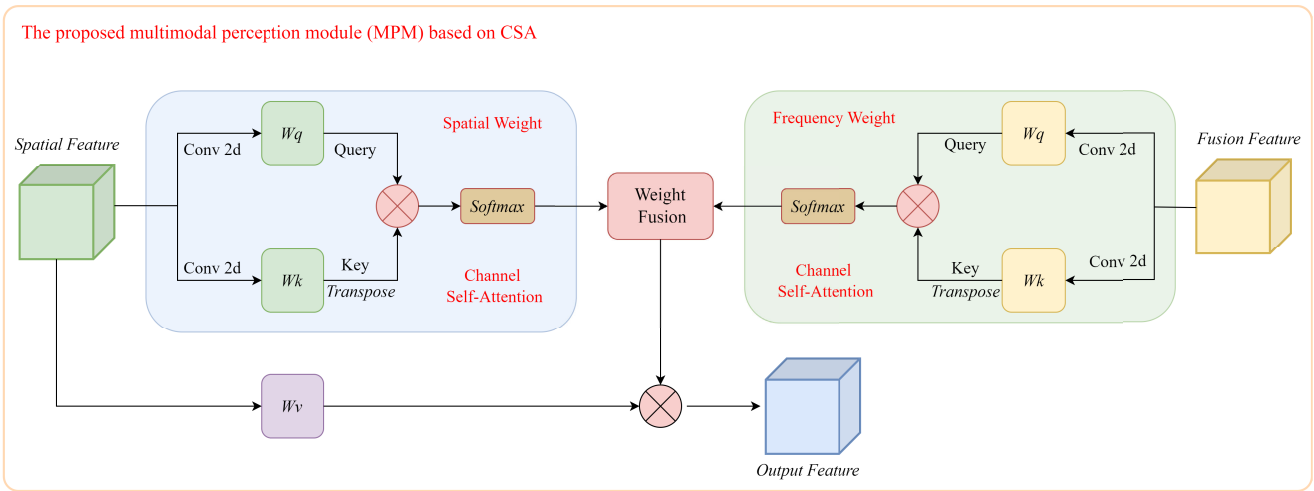


Fig. 7. SFPM based on CSA.

to compute mAP on the HRSC2016 dataset, maintaining consistency with the baseline model. Before obtaining the mAP value, it is essential to calculate the intersection over union (IoU) between the predicted bounding box and the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:22:32 UTC from IEEE Xplore. Restrictions apply.

TABLE I
ABLATION EXPERIMENTS ON DOTA DATASET USING S^2ANET AS THE BASELINE

DOTA	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
S^2ANet	89.21	79.68	50.75	71.15	**79.31**	77.02	86.86	90.88	82.8	**85.12**	**62.05**	63.6	**66.36**	66.46	53.56	73.67
+CA-FFM	**89.27**	76.85	47.89	69.99	78.29	74.98	86.04	90.86	83.26	84.95	50.45	63.77	65.46	64.49	42.60	71.28
+SFPM	89.25	**81.09**	51.27	71.37	78.95	**77.79**	**86.96**	90.89	**85.70**	84.53	60.36	63.95	66.14	68.72	54.79	74.12
MSFN(Ours)	88.90	79.72	50.16	**73.67**	79.08	77.64	86.66	**90.90**	84.68	84.56	58.87	**66.09**	66.36	**69.22**	**60.12**	**74.45**(+0.78)

ground truth. Below is a definition of the IoU

$$\text{IoU} = \frac{\text{area}(A \cap B)}{\text{area}(A \cup B)} \tag{10}$$

where A denotes the predicted bounding box and B represents the ground truth. If the IoU value exceeds the threshold of 0.5, we consider the object as successfully detected. Based on this, if the object category is accurately classified, we classify this as a true positive (TP). Conversely, an incorrect object prediction is labeled as false positive (FP). Additionally, false negative (FN) signifies the number of positive objects missed during detection. Utilizing TP, FP, and FN, we can compute precision (P) and recall (R) using the following equation:

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{11}$$

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{12}$$

The model's detection performance is assessed through both precision and recall. Average precision (AP) offers a comprehensive evaluation by computing the area under the precision–recall (PR) curve. The mathematical expression for AP is as follows:

$$\text{AP} = \int_0^1 p(r)dr. \tag{13}$$

In this equation, p represents precision and r stands for recall. However, AP solely evaluates detection accuracy for individual object categories. Many remote sensing datasets contain multiple object types, such as DOTA, which includes 15 categories. To obtain a comprehensive assessment of the final detection performance across all object categories, we compute the mAP by averaging the AP values for each category of objects. The calculation expression for mAP is as follows:

$$\text{mAP} = \frac{1}{N} \sum_{i=1}^{N} \text{AP}_\text{i} \tag{14}$$

where N denotes the number of object categories in the dataset.

C. Implementation Details

Existing oriented object detectors can be categorized into single-stage detectors and two-stage detectors based on whether they require region proposal networks. Among them, S^2ANet [20] is a typical representative of single-stage oriented object detectors, while Oriented R-CNN [4] serves as a quintessential example of two-stage oriented object detectors. We, respectively, take S^2ANet and Oriented R-CNN as the baselines to validate the effectiveness and generality of the proposed method.

We adopt mmrotate as the object detection framework, with both baseline models using ResNet50 as the backbone network. And it is initialized with pretrained weights from ImageNet dataset for better feature extraction. On the DOTA dataset, we train the models with 3090 GPU for 1 schedule, equivalent to 12 epochs. On the HRSC2016 dataset, we train the models with 3090 GPU for 3 schedules, equivalent to 36 epochs. To mitigate the impact of algorithmic randomness on the final detection results, we conduct each experiment three times while keeping the parameters unchanged, and then select the highest precision.

D. Ablation Study

To validate the effectiveness of the proposed modules, we conduct ablation experiments on the DOTA and HRSC2016 datasets using both the single-stage object detector S^2ANet [20] and the two-stage object detector Oriented R-CNN [4]. In these experiments, CA-FFM represents the channel alignment feature fusion module, SFPM represents the SFPM based on CSA, and MSFN is the network, which integrated with both modules. The results of the ablation experiments for the proposed modules are detailed in the following text.

1) Ablation Experiments on the Single-Stage Object Detector Baseline S^2ANet: The ablation experiment results on the DOTA dataset are presented in Table I. It is notable that incorporating solely the CA-FFM resulted in a 2% decrease in the model's detection accuracy when compared to the baseline network. This decline is attributed to the fact that the CA-FFM only performs simple fusion of spatial-frequency features, failing to fully exploit the object orientation and texture information inherent in frequency domain features. When only the SFPM is added to the backbone network, the model's detection accuracy improved by 0.45% compared to the baseline. This indicates that our proposed SFPM can effectively utilize frequency information in channel dimension and perform feature relocation and redistribution, validating the effectiveness of the CSA mechanism. Furthermore, with the integration of both the CA-FFM and SFPM into the backbone network to form the MSFN, the model's detection accuracy experienced a further enhancement compared to the scenario where only the SFPM was incorporated. This suggests that the proposed model MSFN effectively leverages the object orientation and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:22:32 UTC from IEEE Xplore. Restrictions apply.

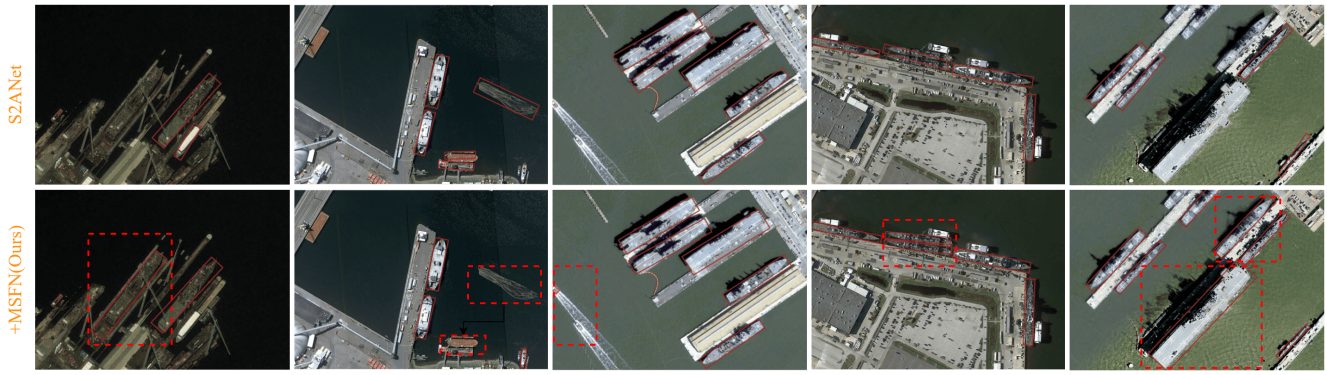


Fig. 8. Visual detection results on HRSC2016 dataset using S^2ANet as the baseline.

TABLE II
ABLATION EXPERIMENTS ON DOTA DATASET USING ORIENTED R-CNN AS THE BASELINE

DOTA	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
Oriented R-CNN	89.15	81.73	53.26	71.22	78.69	82.99	**87.91**	90.88	85.82	84.66	**63.89**	63.62	73.66	**71.02**	55.45	75.35
+CA-FFM	**89.55**	78.93	52.04	71.06	78.34	82.24	87.61	**90.90**	84.81	84.10	57.91	62.19	67.24	65.21	47.22	73.29
+SFPM	89.32	80.48	**54.40**	70.99	78.55	**83.62**	87.87	90.89	**87.09**	85.20	62.36	64.67	**74.44**	70.04	57.79	75.82
+MSFN(Ours)	89.31	**83.66**	53.43	**72.42**	**79.66**	82.50	87.82	90.89	86.27	**85.15**	63.36	**64.80**	73.46	69.29	**60.99**	**76.20**(+0.85)

TABLE III
ABLATION EXPERIMENTS ON HRSC2016 DATASET USING S^2ANET AS THE BASELINE

HRSC	mAP_{50}	mAP_{75}	$mAP_{50,75}$	$mAP_{50\sim95}$
S2ANet	89.6	65.1	77.4	56.0
+CA-FFM	88.9	56.8	72.9	52.7
+SFPM	89.8	57.4	73.6	55.5
+MSFN(Ours)	**89.9**(+0.3)	**67.3**(+2.2)	**78.6**(+1.2)	**56.6**(+0.6)

texture information inherent in frequency domain features and enhances them through the SFPM based on CSA. Finally, the proposed MSFN achieves a 0.78% improvement in object detection accuracy compared to the baseline model. Among the 15 categories in the DOTA dataset, our proposed model achieved superior detection results in 11 categories.

From the experimental results shown in Table III, it is evident that our proposed model exhibits an improvement in detection accuracy on the HRSC2016 dataset. As can be seen from the above table, using mAP50 as the evaluation metric, the model's detection accuracy improved by 0.3%; with mAP75 as the evaluation metric, the model's detection accuracy improved by 2.2%. And when using the mAP(50,95) metric in the COCO dataset format, the model's detection accuracy increased by 0.63%. While the individual addition of the CA-FFM led to a slight decrease in detection accuracy, incorporating the SFPM based on CSA resulted in significant improvements, particularly when evaluated with the mAP75 metric. Furthermore, the combination of the two proposed modules significantly boosted the model's detection accuracy, showcasing the SFPM's effectiveness in leveraging object orientation and texture information extracted by the CA-FFM. The visual detection results of our proposed method and the baseline network S^2ANet on the HRSC2016 dataset are shown in Fig. 8.

2) Ablation Experiments on the Two-Stage Object Detector Baseline Oriented R-CNN: The results of the ablation experiments on the DOTA dataset are presented in Table II. It can be observed that our proposed method, when taking Oriented R-CNN as the baseline, achieves improvements in detection accuracy across all 12 categories in the DOTA dataset. The final detection accuracy, measured by mAP50 metric, increased by 0.85% compared to the baseline network. In the case of solely adding the CA-FFM, there is no improvement in the model's object detection accuracy. However, integrating the SFPM effectively utilizes the frequency domain information brought by CA-FFM, enhancing the model's detection performance on oriented objects in remote sensing images. The experiments demonstrate that the proposed method not only improves single-stage object detection algorithms but also significantly enhances two-stage object detection algorithms, confirming the effectiveness and generality of our proposed method. The visual detection results obtained on the DOTA dataset, using Oriented R-CNN as the baseline, are shown in Fig. 9.

The results of the ablation experiments on the HRSC2016 dataset are shown in Table IV. It can be observed that our proposed method exhibits significant improvement on the HRSC2016 dataset compared to the baseline model Oriented R-CNN. Specifically, when using mAP50 as the evaluation metric, the model's detection accuracy increased by 0.1%. While using mAP75 as the evaluation standard, the model's detection accuracy improved by 0.9%. Additionally, when using the mAP (50 95) metric from the COCO dataset format

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:22:32 UTC from IEEE Xplore. Restrictions apply.

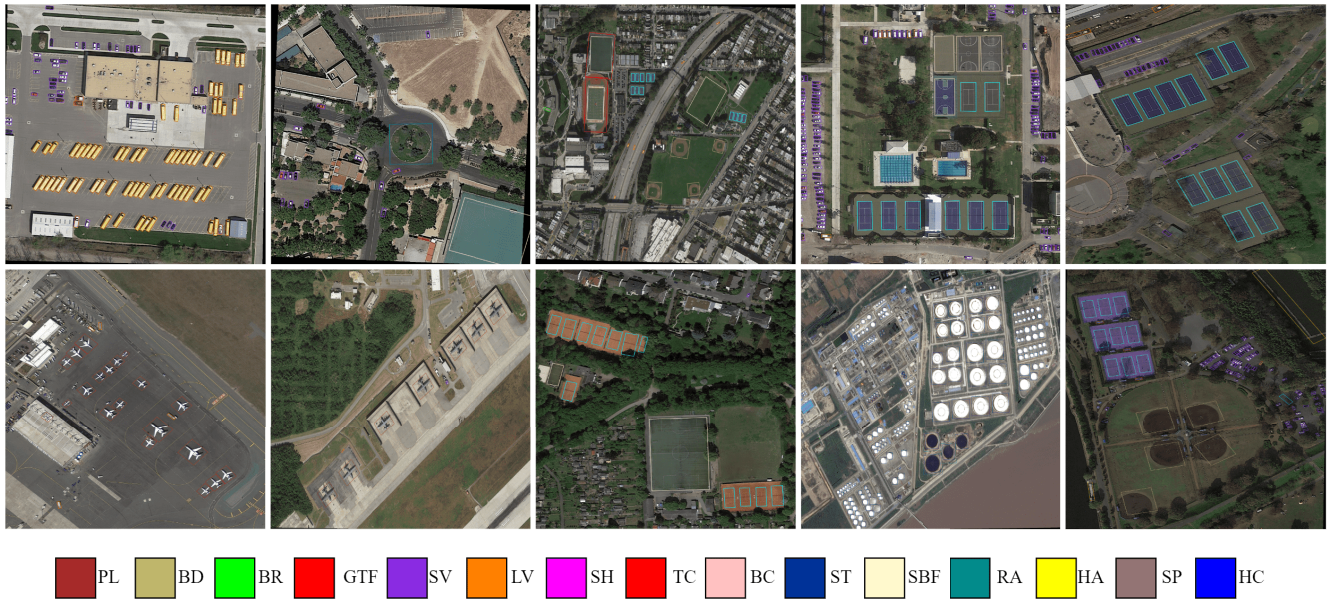


Fig. 9. Visual detection results on DOTA dataset using Oriented R-CNN as the baseline.

TABLE IV
ABLATION EXPERIMENTS ON HRSC2016 DATASET USING ORIENTED R-CNN AS THE BASELINE

HRSC	mAP_{50}	mAP_{75}	$mAP_{50,75}$	$mAP_{50\sim95}$
Oriented R-CNN	90.3	79.6	85.0	65.4
+CA-FFM	89.0	70.1	79.6	60.2
+SFPM	90.4	80.1	85.3	66.5
+MSFN(Ours)	**90.4**(+0.1)	**80.5**(+0.9)	**85.5**(+0.5)	**66.8**(+1.4)

as the evaluation criterion, the model's detection accuracy increased by 1.36%. As the evaluation metric IoU threshold increases, the improvement in detection accuracy of our proposed model compared to the baseline model also increases. The visual detection result between our proposed method and the baseline network Oriented R-CNN on the HRSC2016 dataset is shown in Fig. 10.

The ablation experiment results conducted on the DOTA and HRSC2016 datasets, with S^2ANet and Oriented R-CNN as baselines, demonstrate the effectiveness of our proposed method. Additionally, the results indicate that our method can effectively improve the detection accuracy of oriented objects across various types of object detectors.

E. Comparison With State-of-the-Art Methods

As the proposed module works on the backbone network of the oriented object detector, its objective is to leverage the angle and texture information inherent in the frequency domain features of remote sensing images, thereby enhancing the detection accuracy of oriented objects. It is also designed to be plug-and-play, facilitating its application in various CNN-based object detectors. To validate the effectiveness and universality of the proposed modules, we compare the proposed model MSFN with current mainstream algorithms on the DOTA and HRSC2016 datasets. The original detection accuracies of each comparison methods are reproduced locally. Keeping other parameters unchanged, integrating the plug-and-play modules proposed in this article into the backbone network yields corresponding comparative experimental results. To eliminate the randomness of the experimental results, each comparative method is tested three times, and the highest precision among them is taken as the final experimental result. The selected oriented object detection algorithms for comparative experiments are as follows: RetinaNet [47], R3Det [19], Oriented RepPoints [48], FCOS [49], ATSS [50], SASM [51], KLD [52], RoITrans [2], and Gliding Vertex [25].

1) Comparative Experimental Results on the DOTA Dataset: As can be seen from Table V, it is evident that after incorporating our proposed MSFN, the detection accuracy of different comparative algorithms on the DOTA dataset has been improved to varying degrees, with an average precision increase of over 1%. The quantitative analysis is as follows: when adding the MSFN module to RetinaNet, the model's detection accuracy improved by 1.31%. Compared to R3Det, the detection accuracy increased by 1.18%. Compared to Oriented RepPoints, the detection accuracy increased by 1%. Compared to FCOS, the detection accuracy increased by 0.49%. Compared to ATSS, the detection accuracy increased by 0.31%. Compared to SASM, the detection accuracy increased by 2.42%. Compared to KLD, the detection accuracy increased by 1.17%. Compared to RoITrans, the detection accuracy increased by 0.55%. Compared to Gliding Vertex, the detection accuracy increased by 0.6%. The experimental results on the aforementioned comparative methods fully demonstrate the effectiveness of our proposed module and its ease of integration into various oriented object detection frameworks, thereby improving the accuracy of object detection in remote sensing images. This validates the plug-and-play nature of the proposed module and the universality of using angle and texture information extracted from frequency

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:22:32 UTC from IEEE Xplore. Restrictions apply.



Fig. 10. Visual detection results on HRSC2016 dataset using Oriented R-CNN as the baseline.

TABLE V
RESULTS OF DIFFERENT METHODS ON DOTA DATASET

DOTA	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
RetinaNet	89.64	78.19	40.99	67.82	77.35	61.00	76.98	90.70	81.04	79.25	55.52	64.66	57.19	65.69	40.84	68.46
+MSFN	89.63	**79.49**	40.32	66.73	**77.95**	**63.10**	76.90	90.62	**83.69**	**82.03**	**55.75**	**66.84**	**58.10**	**67.54**	**47.94**	**69.77**(+1.31)
R3Det	89.18	75.78	45.30	69.48	76.10	73.24	79.22	90.89	79.15	83.81	57.76	63.09	63.67	61.90	28.91	69.16
+MSFN	**89.21**	71.88	**46.40**	**70.49**	**77.65**	**74.19**	**81.22**	90.89	**80.52**	**84.31**	**60.39**	62.52	**65.32**	**66.50**	**33.62**	**70.34**(+1.18)
Reppoints	87.93	68.56	51.37	71.78	78.16	76.36	87.27	90.90	81.51	85.69	50.39	62.45	65.01	64.00	26.87	69.88
+MSFN	**88.33**	**74.71**	**51.94**	70.50	**79.19**	**76.87**	**87.53**	90.90	**83.29**	84.92	**52.55**	**62.56**	**65.46**	**65.41**	**29.03**	**70.88**(+1.00)
FCOS	88.66	73.09	47.75	57.19	79.46	76.20	86.35	90.90	83.33	84.63	57.13	65.12	64.18	68.25	37.09	70.62
+MSFN	**88.84**	**74.59**	47.72	**59.98**	78.84	73.42	**86.41**	90.89	**84.07**	84.42	**57.48**	63.20	63.18	68.03	**45.64**	**71.11**(+0.49)
ATSS	88.30	77.82	49.28	70.60	75.63	73.67	81.78	90.90	83.75	84.68	60.88	67.22	64.63	70.30	44.04	72.23
+MSFN	**89.16**	**78.53**	48.64	**71.09**	**77.61**	72.45	79.82	90.77	**85.74**	82.58	60.11	66.94	**64.71**	69.02	**50.87**	**72.54**((+0.31))
SASM	80.56	47.76	46.59	62.00	72.53	72.14	79.81	90.85	71.17	83.43	32.86	54.94	62.21	55.95	33.09	63.05
+MSFN	**85.03**	**61.23**	44.35	**65.79**	**73.87**	**73.07**	**79.85**	**90.86**	**71.61**	83.21	**44.54**	**57.69**	**63.67**	**59.85**	27.44	**65.47**(+2.42)
KLD	89.58	80.20	42.06	69.42	78.31	64.93	82.19	90.88	82.30	82.46	58.52	61.35	58.27	65.45	34.88	69.39
+MSFN	89.33	79.28	39.61	**71.06**	**79.30**	**67.25**	**83.91**	90.88	**82.47**	**82.61**	57.13	**64.70**	55.52	**66.71**	**48.77**	**70.56**(+1.17)
RoITrans	89.11	82.75	52.75	70.81	78.78	87.06	87.69	90.89	87.06	85.86	64.03	62.53	74.52	72.29	57.36	75.61
+MSFN	88.86	82.56	**52.83**	**71.78**	**79.75**	82.42	**87.73**	90.88	86.56	**86.50**	62.33	60.33	**74.98**	72.18	**61.94**	**76.16**(+0.55)
Gliding Vertex	88.96	73.96	50.51	66.77	78.46	75.50	86.34	90.90	79.53	85.59	54.99	63.36	65.96	70.58	55.82	72.48
+MSFN	88.63	**75.63**	**51.14**	**68.36**	78.16	**75.95**	**86.61**	90.89	**83.96**	84.82	**55.65**	**65.56**	**66.61**	68.94	55.35	**73.08**(+0.60)

domain features to enhance the spatial features in remote sensing images.

Table VI presents the comparative experimental results obtained on the DOTA dataset using different evaluation metrics, where, mAP50 represents the mAP with an IoU threshold of 0.5, mAP75 represents the mAP with an IoU threshold of 0.75, mAP(50,95) represents the mean mAP over IoU thresholds from 0.5 to 0.95 in COCO format. The experimental results demonstrate comprehensively that our proposed method can improve the detection accuracy of oriented objects across various metrics, validating the effectiveness of the model.

2) Comparative Experimental Results on the HRSC2016 Dataset: Table VII shows the comparative experimental results on the HRSC2016 dataset, using four different IoU thresholds. It can be seen from the table that our proposed MSFN improves the object detection accuracy over the aforementioned nine comparative algorithms when evaluated based on mAP50. However, for R3Det, KLD, and Gliding Vertex, a slight decrease in detection accuracy is observed when using mAP75 and mAP(50,95) as evaluation criteria. Conversely, our proposed model demonstrates significant improvements in detection accuracy across the remaining six algorithms when evaluated based on mAP75 and mAP(50,95). Specifically, when evaluating detection accuracy on the RetinaNet model using mAP75, there is a significant improvement, reaching 8.2%. Similarly, on the Oriented RepPoints model, there is an 8.3% improvement when using mAP75. Furthermore, for FCOS, ATSS, and SASM, when using mAP75, the detection accuracy improves by 2.5%, 1.6%, and 2.4%, respectively.

Overall, as observed from Tables VI and VII, the proposed model exhibits limited improvement when evaluated against Gliding Vertex as the baseline on the DOTA and HRSC2016 datasets. The detection accuracy under the mAP50 standard shows increases of 0.6% and 0.9%, respectively, while a decline is observed under the mAP75 standard. However, significant enhancements in detection accuracy are achieved

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:22:32 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
RESULTS OF DIFFERENT METHODS ON DOTA DATASET WITH DIFFERENT IoU THRESHOLD

DOTA	mAP_{50}	mAP_{75}	$mAP_{50,75}$	$mAP_{50\sim95}$
RetinaNet	68.46	36.75	52.61	38.54
+MSFN	69.77(+1.31)	38.09(+1.34)	53.93(+1.32)	39.44(+0.9)
R3Det	69.16	36.03	52.59	37.53
+MSFN	70.34(+1.18)	36.33(+0.3)	53.33(+0.74)	38.02(+0.49)
Reppoints	69.88	41.36	55.62	40.60
+MSFN	70.88(+1.00)	42.26(+0.90)	56.57(+0.95)	41.56(+0.96)
FCOS	70.62	39.36	54.99	40.10
+MSFN	71.11(+0.49)	37.88(-1.48)	54.50(-0.49)	39.56(-0.54)
ATSS	72.23	37.61	54.29	40.04
+MSFN	72.54(+0.31)	38.99(+1.38)	55.76(+1.47)	40.85(+0.81)
SASM	63.05	34.97	49.01	35.54
+MSFN	65.47(+2.42)	37.02(+2.05)	51.24(+2.23)	37.18(+1.64)
KLD	69.39	38.69	54.04	39.46
+MSFN	70.56(+1.17)	39.91(+1.21)	55.23(+1.19)	40.09(+0.63)
RoITrans	75.61	47.54	61.57	45.88
+MSFN	76.16(+0.55)	45.97(-1.78)	61.07(-0.50)	45.28(-0.60)
Gliding Vertex	72.48	37.09	54.78	38.89
+MSFN	73.08(+0.60)	35.97(-1.12)	54.52(-0.26)	38.91(+0.02)

TABLE VII
RESULTS OF DIFFERENT METHODS ON HRSC2016 DATASET WITH DIFFERENT IoU THRESHOLD

HRSC	mAP_{50}	mAP_{75}	$mAP_{50,75}$	$mAP_{50\sim95}$
RetinaNet	84.4	49.5	67.0	47.9
+MSFN	85.4(+1.0)	57.7(+8.2)	71.6(+4.6)	52.6(+4.7)
R3Det	84.0	47.1	65.6	46.9
+MSFN	84.5(+0.5)	45.6(-1.5)	65.2(-0.5)	46.2(-0.7)
Reppoints	86.4	56.4	71.4	52.2
+MSFN	87.8(+1.4)	64.7(+8.3)	76.3(+4.9)	56.2(+4.0)
FCOS	87.8	53.1	70.5	50.6
+MSFN	88.8(+1.0)	55.6(+2.5)	72.2(+1.7)	53.3(+2.7)
ATSS	86.9	64.7	75.8	53.6
+MSFN	87.4(+0.5)	66.3(+1.6)	76.9(+1.1)	55.1(+1.5)
SASM	82.3	51.7	67.0	47.3
+MSFN	82.7(+0.4)	54.1(+2.4)	68.4(+1.4)	49.3(+2.0)
KLD	86.3	59.8	73.1	53.9
+MSFN	87.6(+1.3)	58.4(-1.4)	73.0(-0.1)	53.3(-0.6)
RoITrans	89.1	66.3	77.7	55.2
+MSFN	89.5(+0.4)	66.4(+0.1)	78.0(+0.3)	56.6(+1.0)
Gliding Vertex	79.5	39.0	59.3	41.7
+MSFN	80.4(+0.9)	38.3(-0.7)	59.4(+0.1)	41.2(-0.5)

when using baselines such as RetinaNet, RepPoints, and ATSS. This phenomenon can be attributed to the fact that the proposed MSFN serves as a general backbone network for oriented object detection in remote sensing, which does not guarantee improvements in detection accuracy across all metrics. Nonetheless, its performance enhancement is noteworthy for the majority of baseline models. Therefore, based on this analysis, it can be concluded that the proposed MSFN is effective and state-of-the-art.

V. CONCLUSION

This article addresses the challenge of complex background interference and the difficulty of capturing object orientation information with horizontal convolutional kernels in remote sensing images. We propose a multiscale spatial-frequency domain network (MSFN) to utilize object angle and texture information inherent in object frequency domain features. The proposed MSFN consists of the CA-FFM and SFPM. To address the problem of fusing low-level texture and high-level semantic features with a significant difference in channel dimension, we propose CA-FFM. In order to improve the quality of the fused two modality features, we innovatively design the SFPM based on CSA to reposition and redistribute the fused spatial-frequency domain features along the channel dimension. We calculate channel-wise attention weights in the form of self-attention, providing a novel perspective on the application of self-attention mechanism. Moreover, the proposed MSFN can be easily integrated into CNN backbone in a plug-and-play manner, demonstrating its universality and advancement.

REFERENCES

[1] S. Zheng, Z. Wu, Y. Xu, Z. Wei, and A. Plaza, "Learning orientation information from frequency-domain for oriented object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5628512.
[2] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2849–2858.
[3] J. Ma et al., "Arbitrary-oriented scene text detection via rotation proposals," *IEEE Trans. Multimedia*, vol. 20, no. 11, pp. 3111–3122, Nov. 2018.
[4] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3520–3529.
[5] L. Wang, Y. Zhan, W. Liu, B. Yu, and D. Tao, "Bounding box vectorization for oriented object detection with Tanimoto coefficient regression," *IEEE Trans. Multimedia*, vol. 26, pp. 5181–5193, 2024.
[6] Y. Liang, J. Feng, X. Zhang, J. Zhang, and L. Jiao, "MidNet: An anchor-and-angle-free detector for oriented ship detection in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5612113.
[7] J. Feng, Y. Liang, X. Zhang, J. Zhang, and L. Jiao, "SDANet: Semantic-embedded density adaptive network for moving vehicle detection in satellite videos," *IEEE Trans. Image Process.*, vol. 32, pp. 1788–1801, 2023.
[8] Q. Ming, L. Miao, Z. Zhou, and Y. Dong, "CFC-Net: A critical feature capturing network for arbitrary-oriented object detection in remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5605814.
[9] P. Wang, X. Sun, W. Diao, and K. Fu, "FMSSD: Feature-merged single-shot detection for multiscale objects in large-scale remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3377–3390, Dec. 2020.
[10] Y. Li, Q. Hou, Z. Zheng, M.-M. Cheng, J. Yang, and X. Li, "Large selective kernel network for remote sensing object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 16748–16759.
[11] X. Cai, Q. Lai, Y. Wang, W. Wang, Z. Sun, and Y. Yao, "Poly kernel inception network for remote sensing detection," 2024, *arXiv:2403.06258*.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:22:32 UTC from IEEE Xplore. Restrictions apply.

[12] J. Han, J. Ding, N. Xue, and G. Xia, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2786–2795.

[13] S. Mei, R. Jiang, M. Ma, and C. Song, "Rotation-invariant feature learning via convolutional neural network with cyclic polar coordinates convolutional layer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5600713.

[14] Y. Tian, M. Zhang, J. Li, Y. Li, H. Yang, and W. Li, "FPNFormer: Rethink the method of processing the rotation-invariance and rotation-equivariance on arbitrary-oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5605610.

[15] Y. Yu and F. Da, "Phase-shifting coder: Predicting accurate orientation in oriented object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 13354–13363.

[16] Z. Zhao and S. Li, "ABFL: Angular boundary discontinuity free loss for arbitrary oriented object detection in aerial images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611411.

[17] M. Gong, H. Zhao, Y. Wu, Z. Tang, K.-Y. Feng, and K. Sheng, "Dual appearance-aware enhancement for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5602914.

[18] S. Zheng, Z. Wu, Y. Xu, and Z. Wei, "Instance-aware spatial-frequency feature fusion detector for oriented object detection in remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5606513.

[19] X. Yang, J. Yan, Z. Feng, and T. He, "R3Det: Refined single-stage detector with feature refinement for rotating object," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 3163–3171.

[20] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–11, 2021.

[21] X. Yang et al., "SCRDet: Towards more robust detection for small, cluttered and rotated objects," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2019, pp. 8232–8241.

[22] W. Qian, X. Yang, S. Peng, X. Zhang, and J. Yan, "RSDet++: Point-based modulated loss for more accurate rotated object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 11, pp. 7869–7879, Nov. 2022.

[23] X. Yang and J. Yan, "On the arbitrary-oriented object detection: Classification based approaches revisited," *Int. J. Comput. Vis.*, vol. 130, no. 5, pp. 1340–1365, May 2022.

[24] X. Yang, L. Hou, Y. Zhou, W. Wang, and J. Yan, "Dense label encoding for boundary discontinuity free rotation detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15819–15829.

[25] Y. Xu et al., "Gliding vertex on the horizontal bounding box for multi-oriented object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1452–1459, Apr. 2020.

[26] M. Ehrlich and L. Davis, "Deep residual learning in the JPEG transform domain," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 3484–3493.

[27] Z. Qin, P. Zhang, F. Wu, and X. Li, "FcaNet: Frequency channel attention networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2021, pp. 783–792.

[28] F. Zhang, A. Panahi, and G. Gao, "FsaNet: Frequency self-attention for semantic segmentation," *IEEE Trans. Image Process.*, vol. 32, pp. 4757–4772, 2023.

[29] X. Wu, D. Hong, J. Tian, J. Chanussot, W. Li, and R. Tao, "ORSIm detector: A novel object detection framework in optical remote sensing imagery using spatial-frequency channel features," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 5146–5158, Jul. 2019.

[30] Y. Cao, Y. Wu, M. Li, W. Liang, and X. Hu, "DFAF-Net: A dual-frequency PolSAR image classification network based on frequency-aware attention and adaptive feature fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5224318.

[31] U. Shaham, K. Stanton, H. Li, B. Nadler, R. Basri, and Y. Kluger, "SpectralNet: Spectral clustering using deep neural networks," 2018, *arXiv:1801.01587*.

[32] Y. Rao, W. Zhao, Z. Zhu, J. Lu, and J. Zhou, "Global filter networks for image classification," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 980–993.

[33] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[34] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.

[35] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.

[36] S. Xie, S. Liu, Z. Chen, and Z. Tu, "Attentional ShapeContextNet for point cloud recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 4606–4615.

[37] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 603–612.

[38] X. Li, Z. Zhong, J. Wu, Y. Yang, Z. Lin, and H. Liu, "Expectation-maximization attention networks for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9167–9176.

[39] L. Huang, Y. Yuan, J. Guo, C. Zhang, X. Chen, and J. Wang, "Interlaced sparse self-attention for semantic segmentation," 2019, *arXiv:1907.12273*.

[40] Z. Zhu, M. Xu, S. Bai, T. Huang, and X. Bai, "Asymmetric non-local neural networks for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 593–602.

[41] F. Zhang et al., "ACFNet: Attentional class feature network for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6798–6807.

[42] Y. Yuan, X. Chen, and J. Wang, "Object-contextual representations for semantic segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Glasgow, U.K. Cham, Switzerland: Springer, 2020, pp. 173–190.

[43] R. Li, J. Su, C. Duan, and S. Zheng, "Linear attention mechanism: An efficient attention for semantic segmentation," 2020, *arXiv:2007.14902*.

[44] Y. Chen, Y. Kalantidis, J. Li, S. Yan, and J. Feng, "Aˆ2-Nets: Double attention networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 350–359.

[45] S. Wang, Z. Cai, and J. Yuan, "Automatic SAR ship detection based on multifeature fusion network in spatial and frequency domains," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4102111.

[46] H. Zhang, I. Goodfellow, D. Metaxas, and A. Odena, "Self-attention generative adversarial networks," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 7354–7363.

[47] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[48] W. Li, Y. Chen, K. Hu, and J. Zhu, "Oriented reppoints for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1829–1838.

[49] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9627–9636.

[50] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li, "Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9759–9768.

[51] L. Hou, K. Lu, J. Xue, and Y. Li, "Shape-adaptive selection and measurement for oriented object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2022, vol. 36, no. 1, pp. 923–932.

[52] X. Yang et al., "Learning high-precision bounding box for rotated object detection via kullback-leibler divergence," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 18381–18394.



Yang Xu (Member, IEEE) received the B.Sc. degree in applied mathematics and the Ph.D. degree in pattern recognition and intelligence systems from Nanjing University of Science and Technology (NJUST), Nanjing, China, in 2011 and 2016, respectively.

He is currently an Associate Professor with the School of Computer Science and Engineering, NJUST. His research interests include hyperspectral image classification, image processing, and machine learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:22:32 UTC from IEEE Xplore. Restrictions apply.



Yushan Pan (Graduate Student Member, IEEE) was born in Anhui, China, in 2000. He received the B.Sc. degree in computer science and technology from Nanjing University of Science and Technology (NJUST), Nanjing, China, in 2022.

His research interests include object detection and deep learning.



Zebin Wu (Senior Member, IEEE) received the B.Sc. and Ph.D. degrees in computer science and technology from Nanjing University of Science and Technology, Nanjing, China, in 2003 and 2007, respectively.

He was a Visiting Scholar with the Hyperspectral Computing Laboratory, Department of Technology of Computers and Communications, Escuela Politécnica Nacional, University of Extremadura, Cáceres, Spain, from June 2014 to June 2015. He was a Visiting Scholar with the Department of Mathematics, University of California at Los Angeles, Los Angeles, CA, USA, from August 2016 to September 2016 and from July 2017 to August 2017. He was a Visiting Scholar with the GIPSA-Laboratory, Grenoble Institute of Technology (Grenoble INP), Université Grenoble Alpes, Grenoble, France, from August 2018 to September 2018. He is currently a Professor with the School of Computer Science and Engineering, Nanjing University of Science and Technology. His research interests include hyperspectral image processing, parallel computing, and big data processing.



Zhihui Wei (Member, IEEE) was born in Jiangsu, China, in 1963. He received the B.Sc. and M.Sc. degrees in applied mathematics and the Ph.D. degree in communication and information systems from Southeast University, Nanjing, China, in 1983, 1986, and 2003, respectively.

He is currently a Professor and a Ph.D. Supervisor with Nanjing University of Science and Technology (NJUST), Nanjing. His research interests include partial differential equations, mathematical image processing, multiscale analysis, sparse representation, and compressive sensing.



Tianming Zhan received the B.S. and M.S. degrees in applied mathematics from the School of Math and Statistics, Nanjing University of Information Science and Technology, Nanjing, China, in 2006 and 2009, respectively, and the Ph.D. degree in pattern recognition and intelligence systems from the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, in 2013.

He is currently a Professor with the School of Computer and Jiangsu Modern Intelligent Audit Integrated Application Technology Engineering Research Center, Nanjing Audit University, Nanjing. His research interests include hyperspectral image processing, machine learning, and data analysis.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:22:32 UTC from IEEE Xplore. Restrictions apply.